

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2016-2017

MATHEMATICS FOR COMPUTER SCIENTISTS

Time allowed ONE hour and THIRTY minutes only

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all the THREE questions.

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Questions on Number theory, Logic, Sets, and Functions.

(a) For a non-empty tuple of positive integers $a_1, a_2, \dots, a_n$, let

$$\text{CD}(a_1, a_2, \dots, a_n) = \{d \in \mathbb{N} : \forall 1 \leq i \leq n, d|a_i\}$$

be the set of natural numbers that are common divisors of all $a_1, a_2, \dots, a_n$.(i) Using the Fundamental Theorem of Arithmetic (FTA), find $\text{CD}(420, 200, 315)$.

[4 marks]

(ii) Prove or give a counter example that for positive integers a_1, a_2, a_3 , if $\text{CD}(a_1, a_2, a_3) = \{1\}$, then $\text{CD}(a_1 \cdot a_2, a_3) = \{1\}$, where $a_1 \cdot a_2$ denotes the multiplication of a_1 and a_2 .

[6 marks]

(iii) Without using the FTA, prove that for positive integers a and a' , if $\text{CD}(a, a') = \{1\}$ then, for all integers k ,

$$(a \cdot a')|k \leftrightarrow a|k \wedge a'|k$$

[7 marks]

(b) Let $X_n = \{p_1 = 2, p_2 = 3, p_3 = 5, \dots, p_n\}$ be the set of the first n prime numbers. For example, $X_5 = \{2, 3, 5, 7, 11\}$. Let $S_n^{(m)} = \{a \in \mathbb{N} : \forall 0 \leq i \leq n, \exists 0 \leq r_i \leq m \in \mathbb{N}, p_i \in X_n, a = p_1^{r_1} \cdot p_2^{r_2} \cdot \dots \cdot p_i^{r_i} \cdot \dots \cdot p_n^{r_n}\}$ be the set of integers that can be factorised into the multiplication of the first n prime numbers, where the multiplicity of each prime number is less than or equal to a positive integer m .(i) Given an example of the integer $a \in S_5^3$ and explain why it is valid.

[4 marks]

(ii) Specify the cardinality of $S_n^{(m)}$. Show your work.

[4 marks]

(iii) Let $\mathcal{P}(X_n)$ denote the power set of X_n . Define the function $f: S_n^{(m)} \rightarrow \mathcal{P}(X_n)$ such that

$$f(a) = \{p_i \in X_n : \forall 0 \leq i \leq n, r_i > 0, a = p_1^{r_1} \cdot p_2^{r_2} \cdot \dots \cdot p_n^{r_n} \in S_n^{(m)}\}.$$

Show that f is surjective (i.e., onto) but not injective (i.e., not one-to-one).

[9 marks]

2. Questions on Logic, Sets, Induction, and Relations.

(a) A set S is defined recursively by

Basic step: $0 \in S$

Recursive step: if $a \in S$, then $a + 3 \in S$ and $a + 5 \in S$.

(i) Determine the set $S \cap \{a \in \mathbb{Z}: 0 < a < 12\}$.

[5 marks]

(ii) Prove that every integer $a \geq 8$ is contained in S .

[8 marks]

(b) Let $X = \{x, y\}$. Define a binary relation σ on $\mathcal{P}(X)$ the power set of X , by $A \sigma B$ if and only if $A \subseteq B$.

(i) Specify $X \times X$ and $|\mathcal{P}(X)|$.

[4 marks]

(ii) How many different binary relations can be defined on the set X ? Show your work.

[4 marks]

(iii) Is σ reflexive, antisymmetric, and/or transitive? Explain your answer.

[6 marks]

(iv) Represent the relation σ as a Boolean matrix $\mathbf{M}_\sigma$, and determine $\mathbf{M}_\sigma^2$. Show your work.

[6 marks]

3. Questions on Sets, Counting, and Probability.

(a) Suppose that A and B are events from a sample space S . Denote the probability of A and B as $p(A)$ and $p(B)$ respectively.

- (i) Let $S = \{s_1, s_2, s_3, s_4, s_5\}$, $A = \{s_1, s_2, s_4\}$, and $B = \{s_2, s_5\}$. Assuming each sample $s_i, i = 1, \dots, 4$ is equally likely but s_5 happens as twice likely as the rest of the samples. Determine the values of $p(A)$ and $p(B)$ respectively. Justify your answer.

[5 marks]

- (ii) Under the same conditions given in (i), determine the conditional probability $p(A|B)$. Justify your answer.

[4 marks]

- (iii) Given that $p(A) \neq 0$ and $p(B) \neq 0$, show that if $p(B|A) < p(B)$, then $p(A|B) < p(A)$.

[7 marks]

(b) A committee of five is to be chosen randomly from a collection of 4 women and 8 men.

- (i) How many different committees of five people can be formed? Justify your answer.

[5 marks]

- (ii) How many different committees of five can be formed, if at least two men and at least two women are to be on the committee of five? Justify your answer.

[8 marks]

- (iii) What is the probability that there are exactly 3 women and two men chosen, given that the constraints in part (ii) hold?

[4 marks]